Summary in Graph

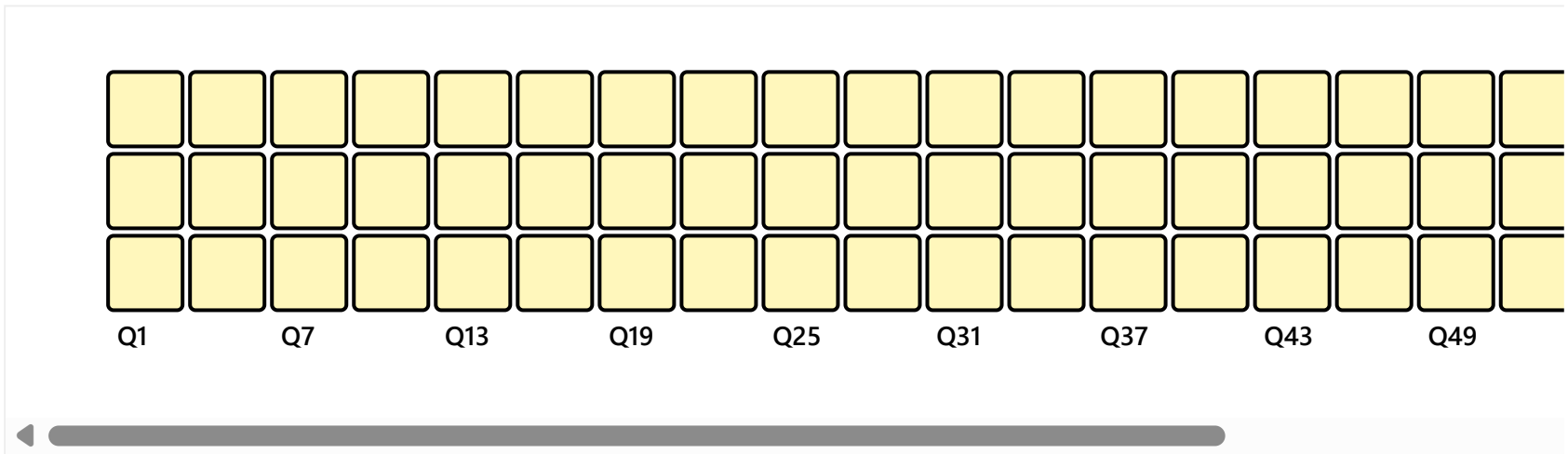
Exam Summary (GO Classes CS Test Series 2026 | MOCK GATE | Test 11)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	65 (30 + 35)
Total Marks:	100 (30 + 70)
Exam Duration:	180 Minutes
Time Taken:	1 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (81) out of 806 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Aptitude

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

Six bells commence tolling together and toll at interval of $2, 4, 6, 8, 10$ and 12 seconds respectively. In 30 minutes, how many times do they toll together?

A. 15
B. 16
C. 20

D. 13

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Others

Tallulah is playing a game in which she rolls a single standard die once. If the number rolled is $1, 2, 3$, or 4 , then she wins. If the number rolled is 5 or 6 , then she loses. What is the probability that she loses?

A. $\frac{1}{3}$
B. $\frac{5}{6}$
C. $\frac{1}{2}$
D. $\frac{1}{6}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Spatial Aptitude

A piece of paper is folded in half, creating two layers of paper. The paper is then folded in half again. This is continued until the paper has been folded in half a total of five times. The total number of layers of paper in the folded sheet is

A. 16
B. 32
C. 25
D. 8

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

In the 3×3 table shown, the numbers 1, 2 and 3 are placed so that each number occurs only once in each row and only once in each column.

		1
3	X	
		Y

The value of $X + Y$ is

A. 2
B. 5
C. 6
D. 4

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

Choose the most appropriate word from the options given below to complete the following sentence:

It was her view that the country's problem had been _____ by foreign technocrats, so that to invite them to come back would be counter-productive.

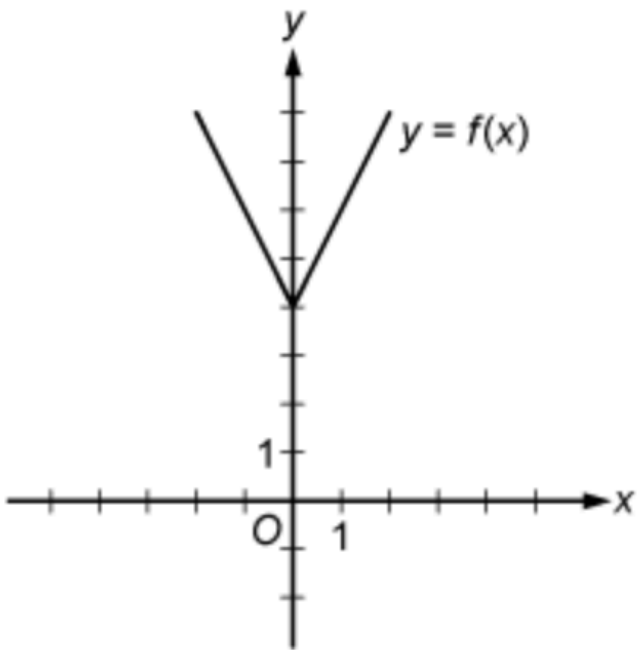
A. identified
B. ascertained
C. exacerbated
D. analysed

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Spatial Aptitude

The figure below shows the graph of the function f defined by $f(x) = |2x| + 4$, for all numbers x.

(Note : $|2x|$ is the absolute value of $2x$)



For which of the following functions g defined for all numbers x does the graph of g intersect the graph of f?

A. $g(x) = x - 2$
B. $g(x) = x + 3$
C. $g(x) = 2x + 3$
D. $g(x) = 3x - 2$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Others

The ratio of the number of quarters (worth $\$0.25$ each) to the number of dimes (worth $\$0.10$ each) to the number of nickels (worth $\$0.05$ each) in a jar is $9 : 3 : 2$. If the total value of the quarters and dimes is $\$17.85$, what is the total value of the nickels?

A. $\$0.45$
B. $\$0.50$
C. $\$0.70$
D. $\$0.35$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

Nine students are sitting side by side in one row in the library while their teacher conducts an online lesson from her home. The teacher sees the class from her laptop screen as shown.

Each student is using a different computer, but the teacher's screen shows who each student is sitting next to. Which student is sitting in the middle (5th position) of the row in the library?

A. Raul
B. Lee
C. Busara
D. Hannah

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 33sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Analytical Aptitude

Zebadiah has 3 red shirts, 3 blue shirts, and 3 green shirts in a drawer. Without looking, he randomly pulls shirts from his drawer one at a time. He would like a set of shirts that includes either 3 of the same colour or 3 of different colours. What is the minimum number of shirts that Zebadiah has to pull out to guarantee that he has such a set?

A. 4
B. 3
C. 6
D. 5

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Verbal Aptitude

Sequence the following sentences $(\mathrm{P}, \mathrm{Q}, \mathrm{R}, \mathrm{S})$ in a coherent passage:

P: Shifu's student exclaimed, "Why do you run since the bull is an illusion?"

Q: Shifu said, "Surely my running away from the bull is also an illusion."

R : Shifu once proclaimed that all life is illusion.

S: One day, when a bull gave him chase, Shifu began running for his life.

A. SPRQ
B. SRPQ
C. RSPQ
D. RPQS

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

Assume the shortest Remaining Time First (SRTF) scheduler for the original workload with jobs A, B, and C. Assume a new job D arrives at time 8.5 seconds.

Job Name	Arrival Time (Seconds)	CPU Burst Time (Seconds)
A	0	8
B	2	4
C	5	7

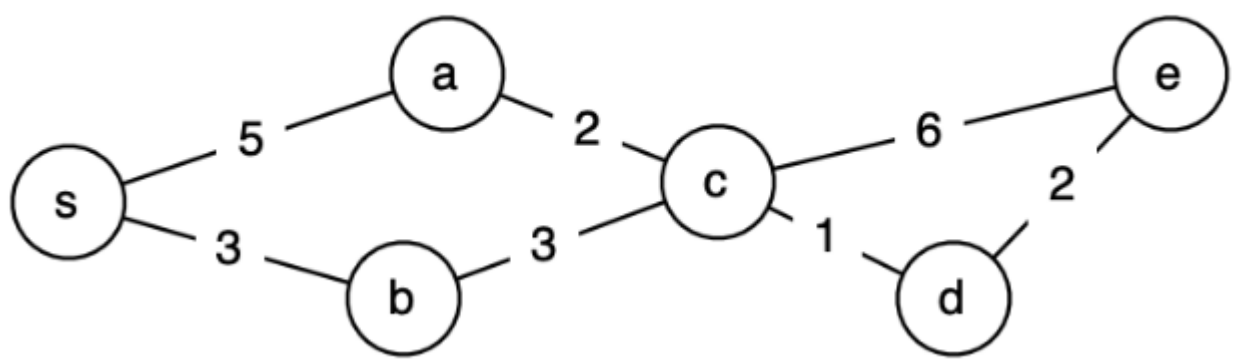
What is the longest that its CPU burst could be to preempt the job that was running at time 8.5 ?

A. 1 seconds
B. 2 seconds
C. 3 seconds
D. 4 seconds

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Numerical Type | Award: 1 | Penalty: 0 | Algorithms

Consider the following weighted graph.



Let X be the length of the shortest path from s to e, and Y be the number of different spanning trees of the graph.
What is the value of $X+Y$?

Your Answer: | Correct Answer: 21 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Suppose we have a sorted list of N unique integers, and we add 2 more integers to the end of the list. What are the exact least and greatest possible number of inversions in the updated list?

A. Least: 0, Greatest: $2N$
B. Least: 1, Greatest: $2N+1$
C. Least: 0, Greatest: $2N+1$
D. Least: 1 , Greatest: $2N$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

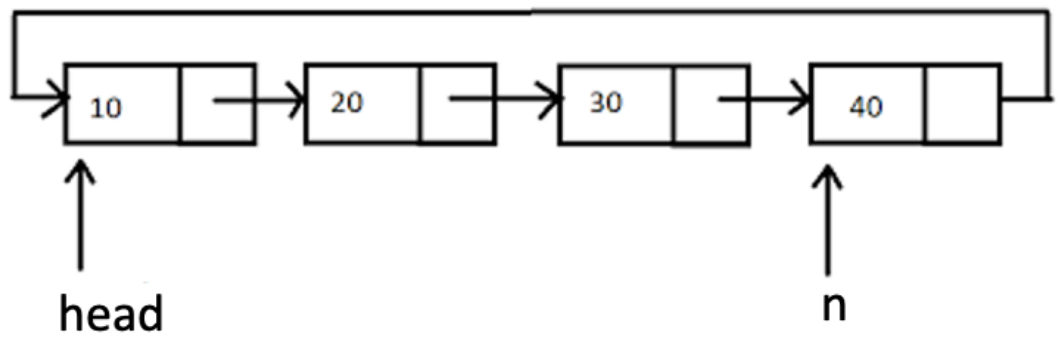
Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider a linked list given in the figure below.

What is the value of n $\rightarrow$ next $\rightarrow$ next $\rightarrow$ next $\rightarrow$ data?

Where n and head are pointers to the following struct type and intitallised as per the figure shown.

```
struct node{
    int data;
    struct node *next;
};
```



A. 10
B. 20
C. 30
D. 40

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Let $\mathrm{C(X)}$ be the number of leaves in a binary tree rooted at T. Assume that $\mathrm{IsLeaf(T)}$ returns 1 if T is a leaf. Which of the following observations leads to a recursive implementation?

A. $\mathrm{C(T)} := \mathrm{C(T.Left)} + \mathrm{C(T.Right)}$
B. $\mathrm{C(T)} := \mathrm{C(T.Left)} + \mathrm{C(T.Right)} + 1$
C. $\mathrm{C(T)} := \mathrm{C(T.Left)} + \mathrm{C(T.Right)} + \mathrm{IsLeaf(T)}$
D. $\mathrm{C(T)} := \mathrm{C(T.Left)} + \mathrm{C(T.Right)} + \mathrm{IsLeaf(T)} + 1$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

Consider that nodes in slotted-Aloha transmit in each slot with probability p. Suppose that a given channel has exactly 8 nodes. What is the probability that there is a collision in a given slot?

A. $1-(1-p)^8$
B. $1-8p(1-p)^7-(1-p)^8$
C. $1-8p(1-p)^7$
D. $(1-p)^8$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

Suppose you are network operator for an ISP, and your ISP has two customer networks $(\mathrm{IITB}\text{ and }\mathrm{IISc})$. For the sake of simplicity, let's assume that IITB only has 3 hosts with the following IP addresses, and IISc only has 2 hosts:

- $\mathrm{IITB}: 18.0.0.1, 18.0.0.254, 18.255.255.254$
- IISc :$128.112.0.1$, and $128.112.255.254$

You have heard that memory in router line cards is expensive, so you would like to use as few entries as possible in your line cards. Your colleague at work says he has a solution that only requires two forwarding-table entries.

Give the IP prefix notation that your colleague uses to represent these two networks.

A. $18.0.0.0/8$ and $128.112.0.0/16$
B. $18.0.0.0/24$ and $128.112.0.0/8$
C. $18.0.0.0/16$ and $128.112.0.0/16$
D. $18.0.0.0/24$ and $128.112.0.0/24$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

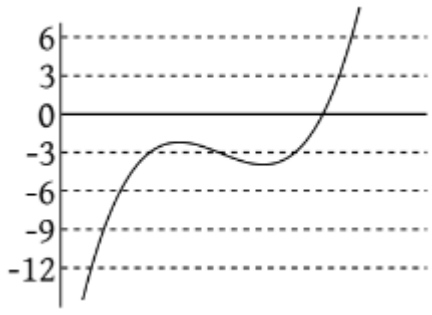
In the Internet Protocol (IP) suite of protocols, which of the following best describes the purpose of the Address Resolution Protocol?

A. To translate Web addresses to host names
B. To determine the IP address of a given host name
C. To determine the hardware address of a given host name
D. To determine the hardware address of a given IP address

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

The graph of a cubic polynomial $f(x)$ is shown below. If k is a constant such that $f(x) = k$ has three real solutions, which of the following could be a possible value of k ?



A. 3
B. 0
C. -7
D. -3

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 1 | Penalty: 0 | Compiler Design

Consider the following simple context-free grammars:

Grammar G_1	Grammar G_2	Grammar G_3
$S \rightarrow A$	$S \rightarrow A$	$S \rightarrow A$
$A \rightarrow \varepsilon$	$A \rightarrow \varepsilon$	$A \rightarrow \varepsilon$
$A \rightarrow bbA$	$A \rightarrow bAb$	$A \rightarrow Abb$

The start symbols are S, the non-terminals are S and A, and b is a terminal symbol. You may also treat $\$$ as a terminal to represent end-of-file. Note that these grammars all generate the same language $= L(G_1) = L(G_2) = L(G_3)$: strings consisting of even numbers of b symbols (including 0 of them).

Which of the following is/are TRUE?

A. G_1 is $LL(1)$
B. G_2 is $LL(1)$
C. G_3 is $LL(1)$
D. Language generated by these grammar is $LL(1)$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in C

What will be printed by following C code?

```
    int a[7] = {0, 1, 2, 3, 4, 5, 6};
    int *p = &a[3];
    p += 2;
    *p += 2;
5. printf("%d", *p++);
```

A. 6
B. 7
C. 8
D. 9

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Find the inverse of the matrix $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

A. $\begin{pmatrix} -1 & 3 \\ 2 & -4 \end{pmatrix}$

B. $\begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix}$

C. $\begin{pmatrix} \frac{1}{2} & -\frac{3}{2} \\ -1 & 2 \end{pmatrix}$

D. $\begin{pmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{pmatrix}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Numerical Type | Award: 1 | Penalty: 0 | Discrete Mathematics

A search committee is formed to find a new software engineer. They will forward to the company CEO a list of four candidates to interview. They will let the CEO know who is their first choice, their second choice, their third choice and their fourth choice. If there are 53 applicants for the job, then how many different outcomes are there from the search committee's selection process?

Your Answer: | Correct Answer: 7027800 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Numerical Type | Award: 1 | Penalty: 0 | Set Theory & Algebra

Given a set of values $\mathrm{R} = \backslash\mathrm{set}1, 2, 3, 4, 5, 6, 7$, What is the number of relations on this set which is both partial-order and equivalence relation?

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Numerical Type | Award: 1 | Penalty: 0 | Mathematical Logic

Consider the following four cards. Suppose each card has a number on one side and a letter on the other side.
Minimum how many cards do we have to turn over to determine whether the following claim is true?
Claim: For each card, if the card has a vowel on one side, then it has an even number on the other side.

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Consider the following schedule:

T1					R(A)		W(C)		R(B)
T2		R(A)	R(B)						
T3	R(A)					W(B)			
T4				W(A)				R(C)	

What is the set of edges of the conflict dependency graph for the schedule? We write (T_i, T_j) if there exists a directed edge from T_i to T_j.

A. $\{(T_1, T_4), (T_3, T_2), (T_4, T_2), (T_1, T_3)\}$
B. $\{(T_2, T_4), (T_3, T_4), (T_4, T_1), (T_2, T_3), (T_1, T_4), (T_3, T_1)\}$
C. $\{(T_4, T_2), (T_4, T_3), (T_1, T_4), (T_3, T_2), (T_4, T_1), (T_1, T_3)\}$
D. None of the above

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Numerical Type | Award: 1 | Penalty: 0 | Databases

Consider the following table named Students in a relational database.

id	gpa	name	address	gender
123	null	Joe	38 Maple	M
124	3.2	Hui	64 Vine	F
127	3.9	Celia	21 Elm	F
111	3.2	Hector	11 Oak	M

The SQL query below is executed on this database.

```
SELECT S.id FROM Students S
 WHERE S.gpa > 3.3 OR S.gender = 'M';
```

How many rows should be in the output?

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Numerical Type | Award: 1 | Penalty: 0 | CO & Architecture

The number of sets in two independent caches C1 and C2 is same. C1 is a 2-way associative cache with 32 byte blocks where as C2 is an 8 -way associative cache with 256 byte blocks. The total storage space of C2 is ____ times larger than C1 .

Your Answer: | Correct Answer: 32 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | CO & Architecture

The content of the registers are $\mathrm{R}_1 = 25\mathrm{H}, \mathrm{R}_2 = 30\mathrm{H}$ and $\mathrm{R}_3 = 40\mathrm{H}$. The following machine instructions are executed.

- $\mathrm{PUSH}\ \{\mathrm{R}_1\}$
- $\mathrm{PUSH}\ \{\mathrm{R}_2\}$
- $\mathrm{PUSH}\ \{\mathrm{R}_3\}$
- $\mathrm{POP}\ \{\mathrm{R}_1\}$
- $\mathrm{POP}\ \{\mathrm{R}_2\}$
- $\mathrm{POP}\ \{\mathrm{R}_3\}$

After execution, the content of registers $\mathrm{R}_1,\ \mathrm{R}_2,\ \mathrm{R}_3$ are

A. $\mathrm{R}_1 = 40\mathrm{H},\ \mathrm{R}_2 = 30\mathrm{H},\ \mathrm{R}_3 = 25\mathrm{H}$
B. $\mathrm{R}_1 = 25\mathrm{H},\ \mathrm{R}_2 = 30\mathrm{H},\ \mathrm{R}_3 = 40\mathrm{H}$
C. $\mathrm{R}_1 = 30\mathrm{H},\ \mathrm{R}_2 = 40\mathrm{H},\ \mathrm{R}_3 = 25\mathrm{H}$
D. $\mathrm{R}_1 = 40\mathrm{H},\ \mathrm{R}_2 = 25\mathrm{H},\ \mathrm{R}_3 = 30\mathrm{H}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Numerical Type | Award: 1 | Penalty: 0 | CO & Architecture

How many interrupts will be required to read 100 bytes of input, one byte at a time, using polling based $\mathrm{I/O}$?

Your Answer: | Correct Answer: 0 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

The IEEE 754 single-precision floating-point representation consists of the following fields:

Sign	Exponent	Mantissa
1 bit	8 bits	23 bits

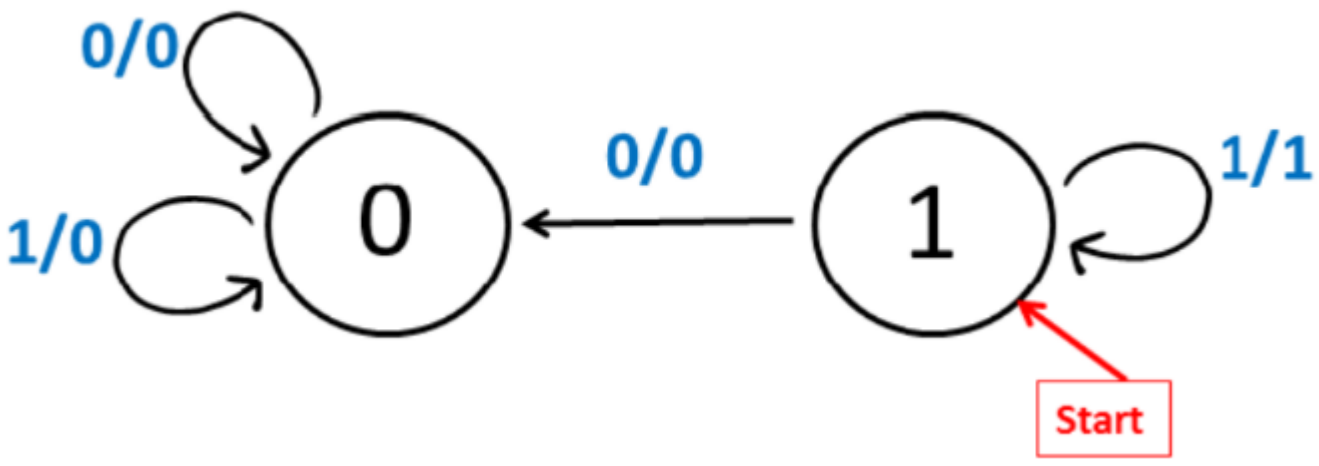
For $x = 2.25$, what will the binary representation of the mantissa be, with the implicit 1 shown?

A. 1.001
B. 0.001
C. 1.010
D. 0.010

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

Consider the following state diagram of a finite state machine:



Which of the following is the correct description of this machine?

A. Each output bit of the finite automaton is the OR of all the input bits seen so far.
B. Each output bit of the finite automaton is the parity of all the input bits seen so far. Recall that the parity of a set of bits is 1 if the number of 1 s is odd and 0 if the number of 1 s is even.
C. Each output bit of the finite automaton is the AND of all the input bits seen so far.
D. Each output bit is the negation of the last input bit seen so far.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Numerical Type | Award: 1 | Penalty: 0 | Digital Logic

In a number system of base r, the equation $x^2 - 12x + 37 = 0$ has $x = 8$ as one of its solutions. The value of r is _____.

Your Answer: | Correct Answer: 11 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

If A is any language, then

A. A^* must be regular.
B. $\bar{A}$ must be regular.
C. $\bar{A} \cup A$ must be regular.
D. none of the above are true.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Select Type | Award: 1 | Penalty: 0 | Theory of Computation

Given the following grammar with nonterminals S, A, and B .
$\mathrm{S} \rightarrow \mathrm{AaBb}$
$\mathrm{A} \rightarrow \mathrm{Ab} \mid \mathrm{b}$
$\mathrm{B} \rightarrow \mathrm{aB} \mid \mathrm{a}$
Which of the following sentences is/are valid ones in the language generated by this grammar?

A. baab
B. bbbab
C. bbaaaa
D. bbaab

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C

Consider the following C code. Where x and y are unknown integers and have been initialized to values greater than 1.

```
    int i = 1, j = 0;
    while (i < y) {
        i = i + 1;
        if (i == x) break;
5. }
```

Which of the following statements about the variable i must be true after the execution of this program?

A. If $x > y$, then $i = y$
B. If $x > y$, then $i = y - 1$
C. If $x > y$, then $i = x$
D. If $x > y$, then $i = x - 1$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in C

Consider the following C code:

```
    int i = 0;
    int j = 0;
    char *s = "abcde";
    while (s[i] != '\0') {
5.          j = j + s[i] - 'a';
            i = i + 1;
    }
    printf("%d %d\n", i, j);
```

What will the program output be?

A. 5 10
B. 5 15
C. 6 10
D. 6 15

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Let $D(n, j)$ be the number of ways to write n as a sum of integers $\{a_1, \ldots, a_j\}$, where each a_i is an integer and used at most once.

For example, if $a_1 = 1, a_2 = 2, a_3 = 3$, and $a_4 = 4$, there are two ways to make $n = 5$ from $\{a_1, a_2, a_3, a_4\} : 5 = 2 + 3$ and $5 = 1 + 4$. Thus, $D(5, 4) = 2$.

Which recurrence relation correctly computes $D(n, k)$?

A. $D(n, j) = D\left(n - a_j, j - 1\right) + D(n, j - 1)$

B. $D(n,j) = D\left(n - a_j, j\right) + D(n, j-1)$
C. $D(n,j) = D(n-1, j) + D(n, j-1)$
D. $D(n,j) = D\left(n - a_j, j-1\right) + D\left(n - a_j, j-1\right)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Given a positive integer d, what is a set of frequencies for 2^d characters where the optimal prefix code has an associated tree of depth exactly d ? (The depth of a tree is the length of the path from the root to the deepest leaf.)

Which of the following options is correct about set of frequencies for 2^d characters ?

A. All characters have the same frequency $\frac{1}{2^d}$.
B. Frequencies follow a geometric progression, with the largest frequency equal to $\frac{1}{2}$ and each subsequent frequency halved.
C. Each frequency is a unique power of $\frac{1}{2}$, summing to 1 .
D. One character has frequency $\frac{1}{2}$, and all other characters share the remaining frequency equally.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Suppose that a binary min-heap stores six elements with priorities $10, 20, 30, 40, 50,$ and 60 in its array A. What is the largest of these items that could be stored in $\mathrm{A}[1]$? (indexing starts from zero)

A. 20
B. 30
C. 40
D. 50

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #31 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

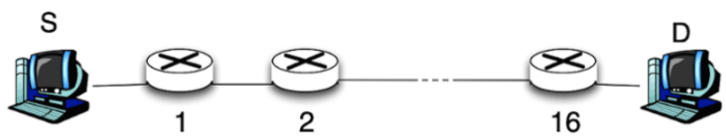
Suppose you wanted to insert the following elements into a binary search tree of integers: $90, 3, 16, 7, 12, 38, 50$. Which of the following insertion orders will result in a *complete binary tree* (where all levels are fully filled except possibly for the last level, which is filled from left to right)*?*

A. $16, 7, 3, 12, 50, 38, 90$
B. $16, 38, 3, 12, 50, 90, 7$
C. $7, 16, 3, 90, 12, 50, 38$
D. $16, 3, 12, 50, 90, 7, 38$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #32 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

In the following network, node S transmits packets that pass through a tandem network of 16 routers, and arrive at the destination D. The bit rate of all links is $\mathrm{R} = 1\ \mathrm{Mbit/sec}$. The maximum packet size in the network is 500 Bytes. Ignore the header size. The one-way propagation delay on each link is $10\ \mathrm{msec}$. Assume that there is no error in transmission in the network, and the size of ACK packets is negligible.



How long does it take to transmit a file of size $50,000$ Bytes if S and D use Stop-and-Wait ARQ?

A. 23.8 seconds
B. 40.8 Seconds
C. 36.7 seconds
D. 7.53 seconds

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #33 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

Ida Investigator sends large UDP datagram with a payload of size 4408 bytes from office machine to her home machine. 4408 bytes is size of application data without UDP header.

She runs a simple listening server on her home machine to observe the traffic. She notices something strange: packet arrives as three IP fragments at her home machine. Each delivered IP fragment is exactly 1500 bytes, including IP headers.

The UDP header is 8 bytes. Calculate the IP header size (in Bytes).

A. 84
B. 20
C. 28
D. 92

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #34 | Multiple Select Type | Award: 2 | Penalty: 0 | Compiler Design

A lexical analyzer uses the following patterns to recognize three tokens T1, T2, and T3 over the alphabet $\{\mathrm{x}, \mathrm{y}\}$:

- T1: x
- T2: xxxx
- T3: $\mathrm{x}^{4n}\mathrm{y}$ for $n \geq 1$

The analyzer outputs the token that matches the longest possible prefix.
For which of the following strings will the lexical analyzer backtrack while tokenizing the string?

A. x
B. $\mathrm{xxxxxxxxy}$ (8 x characters)
C. $\mathrm{xxxxxxxxxxxxxxy}$ (13 x characters)
D. xx

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #35 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Suppose there are n jobs $J_1, J_2, \ldots, J_n$ ready to run, with no other jobs in the system. Each job J_i has a processing time T_i, and the jobs are arranged such that $T_1 < T_2 < \ldots\ldots < T_n$. what order should these jobs be executed in to minimize the average completion time?

A. Jobs should be executed in the order $J_1, J_2, \ldots, J_n$.

B. Jobs should be executed in the order $J_n, J_{n-1}, \ldots, J_1$.

C. Jobs should be executed in the Round Robin order with the time quantum of T_1 units.

D. Any order will result in the same average completion time.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #36 | Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Consider a system with a 6 bit virtual address space, and 16 byte pages/frames. The mapping from virtual page numbers to physical frame numbers of a process is $(0, 8), (1, 3), (2, 11)$, and $(3, 1)$. Note that all addresses are in decimal. Your answer must be in decimal.

Which of the following are is/true?

A. the physical address to which virtual address 20 is mapped to is 52.
B. the physical address to which virtual address 40 is mapped to is 184.
C. The physical address to which virtual address 20 is mapped to is 53.
D. the physical address to which virtual address 40 is mapped to is 180.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #37 | Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Consider the following two threads, to be run concurrently in a shared memory (all variables are shared between the two threads):

Thread A	Thread B
for (i=0 ; i<5 ; i++) { x=x+1; }	for (j=0 ; j<5 ; j++) { x=x+2; }

Assume a single-processor system, that load and store are atomic, that x is initialized to 0 before either thread starts, and that x must be loaded into a register before being incremented (and stored back to memory afterwards). The following options consider the final value of x after both threads have completed.

Which of the following statements about the final value of x is/are true?

A. Final value of x will be lesser or equal to 15.
B. Final value of x can NOT be 1 for any execution order
C. Final value of x can NOT be 2 for any execution order
D. Final value of x is greater than equal to 3.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #38 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

A system is supported with virtual memory and having two processes. Further assume that processes are only reading and not writing. Consider two statements below.

- $\mathrm{S1}$: Both processes reading from the same physical address will access the same contents.
- $\mathrm{S2}$: Both processes reading from the same logical address will access the same contents.

Which of these statements is/are ALWAYS true?

A. Both Statement $\mathrm{S1}$ and Statement $\mathrm{S2}$
B. Only Statement $\mathrm{S1}$
C. Only Statement $\mathrm{S2}$
D. Neither Statement $\mathrm{S1}$ nor Statement $\mathrm{S2}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #39 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Let A be an $\mathrm{n} \times \mathrm{n}$ real matrix for which two distinct non-zero n-dimensional real column vectors v_1 and v_2 satisfy the relation $\mathrm{Av}_1 = \mathrm{Av}_2$. Which of the following statements about A is/are true?

A. At least one eigenvalue of A is zero.

B. A is not full rank.

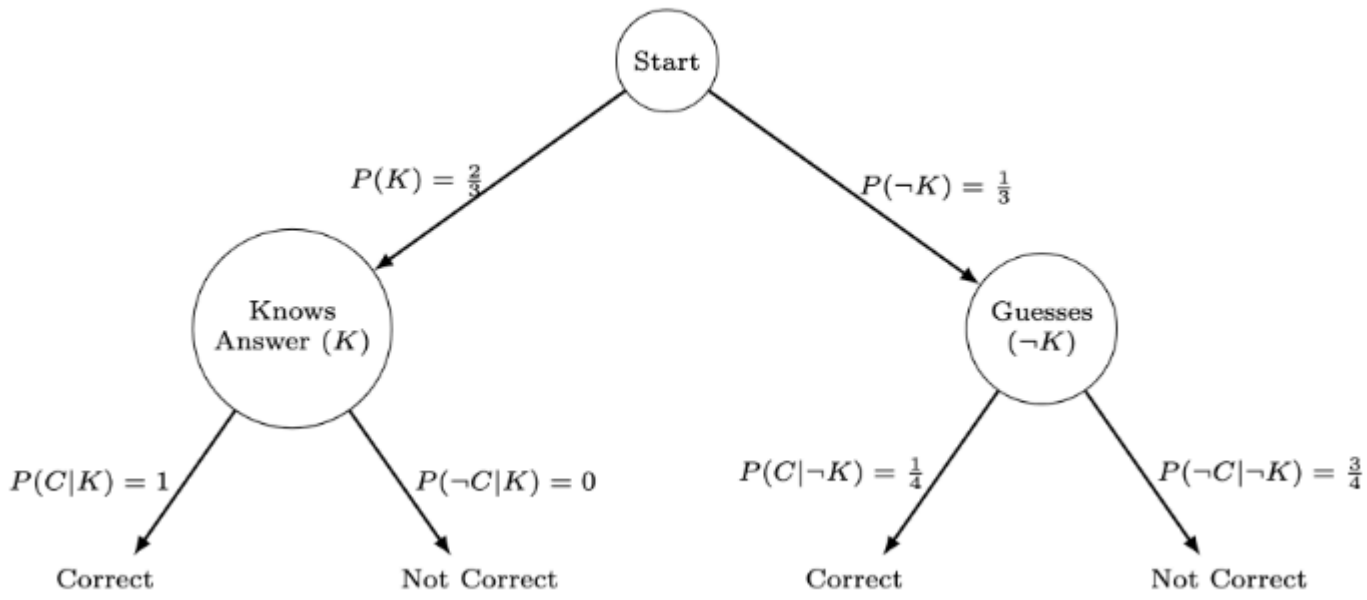
C. Columns of A are linearly independent.

D. The determinant of A is zero.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #40 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

In a multiple-choice test, a student either knows the answer to a question or guesses randomly. Each question has 4 possible answers, and the probability that the student knows the answer to a question is $2/3$.
The following diagram represents this scenario using a tree visualization:



What is the probability that the student knows the answer to a question, given that the answer was correct?

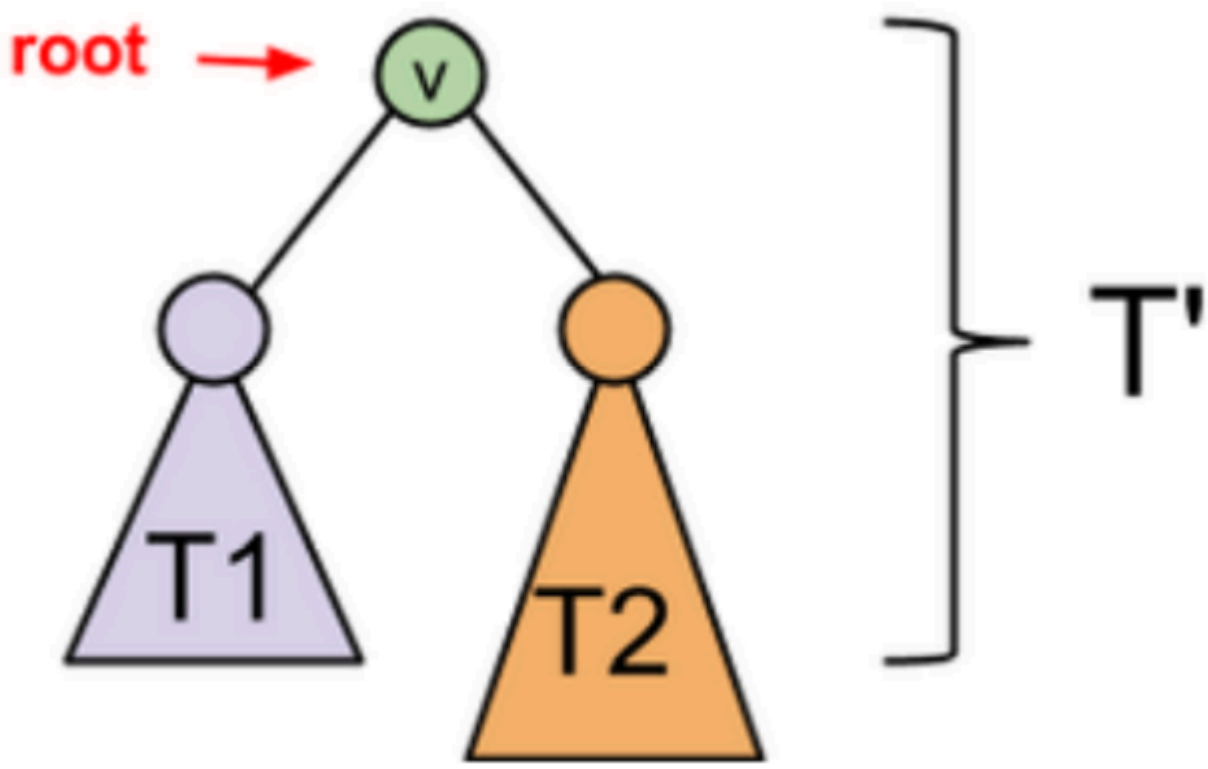
A. $1/2$
B. $2/3$
C. $3/4$
D. $8/9$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

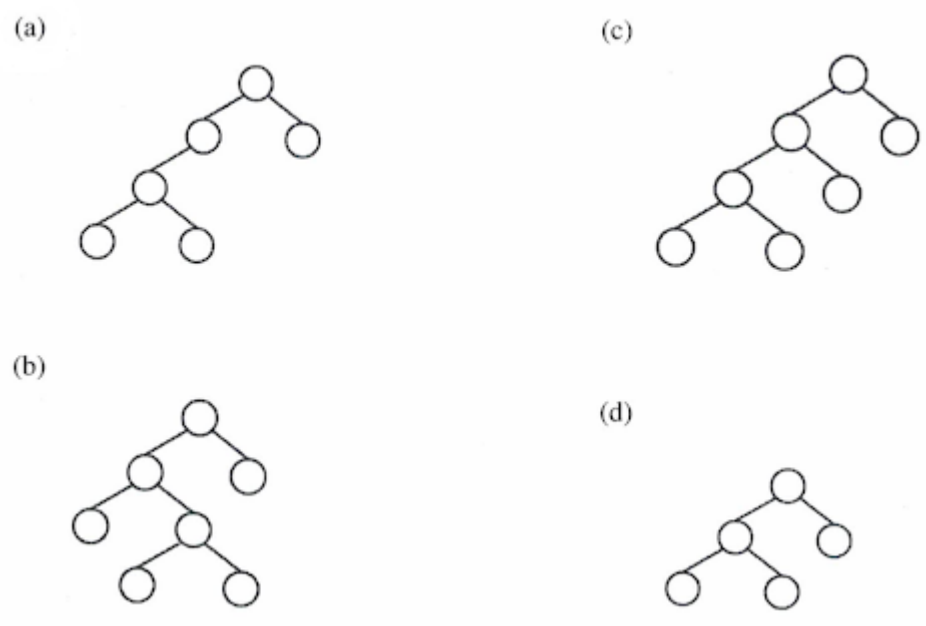
Q #41 | Multiple Select Type | Award: 2 | Penalty: 0 | Graph Theory

Here is a definition for a set of trees called GO trees.
Basis: A single vertex with no edges is a GO tree. The root is the only vertex in the tree.

Recursive rule: If $\mathrm{T1}$ and $\mathrm{T2}$ are GO trees, then a new GO tree T' can be constructed by first placing $\mathrm{T1}$ to the left of $\mathrm{T2}$, adding a new vertex v at the top and then adding an edge between v and the root of $\mathrm{T1}$ and an edge between v and the root of $\mathrm{T2}$. The new vertex v is the root of T'.



Exclusion statement: T is a GO tree only if T is the tree described in the Basis or T can be constructed by repeatedly applying the recursive rule starting with the tree given in the basis.
Which of the following trees is/are NOT a GO tree?



Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #42 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Mathematical Logic

The function receives two inputs: a and n. a is a real number and n is an integer such that $n \geq 1$. It should return

$$\text{SuperPower}(a, n) = a^{4n-2}$$

Note that in the expression above, the exponent of a is $4n-2$. Below is a recursive algorithm to compute SuperPower(a, n) with some lines missing.

```
    SuperPower(a, n)
        If ( A ) Return ( B)  // Base Case
        y := SuperPower( C, D ) // Recursive Call
        return ( E) // Mathematic expression using y and/or a
5. End
```

For the recursive algorithm, what expression should go in the spaces labeled $\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}, \mathrm{E}$?

A. A: $\mathrm{a}=1,\ \mathrm{B}$: a^2, C: $a-1, \mathrm{D}: \mathrm{n}-1, \mathrm{E}: \mathrm{a}.\mathrm{y}^4$
B. A: $\mathrm{n}=1,\ \mathrm{B}: \mathrm{a}^2, \mathrm{C}: \mathrm{a}, \mathrm{D}: \mathrm{n}-1, \mathrm{E}: \mathrm{y}.\mathrm{a}^4$
C. A: $\mathrm{n}=1,\ \mathrm{B}: \mathrm{a}^2, \mathrm{C}: \mathrm{a}, \mathrm{D}: \mathrm{n}-4, \mathrm{E}: \mathrm{y}.\mathrm{a}^4$
D. A: $\mathrm{n}=1,\ \mathrm{B}: \mathrm{a}^2, \mathrm{C}: \mathrm{a}, \mathrm{D}: \mathrm{n}-1, \mathrm{E}: \mathrm{a} \cdot y^4$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #43 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Mathematical Logic

Let $n \geq 2$. Consider the complete graph K_n on n vertices. Color each edge of this graph K_n, independently, as either red or blue. Call the new colored graph as G.

Let R be the spanning subgraph of G consisting of all the red edges only, and let B be the spanning subgraph of G consisting of all the blue edges only.

Then which of the following is true?

A. R is disconnected or B is disconnected.
B. R is disconnected and B is disconnected.
C. R is connected or B is connected.
D. R is connected and B is connected.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #44 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Assume that you are considering a new normal form TANF (Totally Awesome Normal Form). A constraint satisfies TANF for a relation R if, for every functional dependency $\mathrm{X} \rightarrow \mathrm{Y}$, one of the following is true:
i) $\mathrm{X} \rightarrow \mathrm{Y}$ is a trivial FD
ii) X is a candidate key for R
Which of the following is/are true?

A. Every TANF schema is in BCNF.
B. Every BCNF schema is in TANF.
C. For every relation R, there is always a decomposition into TANF that is lossless.
D. For every relation R, there is always a decomposition into TANF that is lossless and dependency preserving.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #45 | Numerical Type | Award: 2 | Penalty: 0 | CO & Architecture

Assuming the following assembly instructions are executed in the order that they are found in the table below, on the standard 5-stage pipeline having the stages:
$\mathrm{Instruction\ Fetch(F)}$, $\mathrm{Decode\ \&\ Read\ Register(D)}$, $\mathrm{Execute(X)}$, $\mathrm{Data\ Memory\ Access(M)}$, $\mathrm{Register\ Write(W)}$ Assume that register writes occur in the first half of the clock cycle and reads in the second half.

Instructions
sub $2, $3, $1
lw $5, 0($2)
addi $4, $5, 1
add $5, $3, $1

In all these assembly instructions, the leftmost register is the destination register for the respective operation. With the pipeline with full bypassing, the code fragment takes X cycles. With the pipeline with no bypassing, the code fragment takes Y cycles. What is Y/X (up to one decimal place)?

Your Answer: | Correct Answer: 1.3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #46 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

Consider two different enhancements to the design of a microprocessor.
The first enhancement $(\mathrm{E1})$ speeds up 5% of the program execution time by $10x$. The second enhancement $(\mathrm{E2})$ speeds up 10% of the program execution time by $5x$. Which optimization provides higher performance improvement? (Mark the correct answer).

A. $\mathrm{E1}$ provides more improvement than $\mathrm{E2}$
B. $\mathrm{E2}$ provides more improvement than $\mathrm{E1}$
C. Both $\mathrm{E1}$ and $\mathrm{E2}$ provide similar improvements
D. Problem statement does not have enough data to conclude which option is better

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #47 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | CO & Architecture

In a computer with a cache memory interposed between the processor and the primary memory, the cache is k-way set-associative (for some fixed k); i.e., each location in primary memory "maps to" (can be cached in) any of k locations in the cache. Let there be P locations in primary memory and C locations in the cache. On the average, how many different locations in the primary memory map to a particular location in the cache if $k=2$?

A. 1
B. 2
C. P / C
D. $2 \mathrm{P} / \mathrm{C}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #48 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

For the Boolean function

$$F(A, B, C, D)=\sum m(0,2,5,7,8,10,12,13,14,15),$$

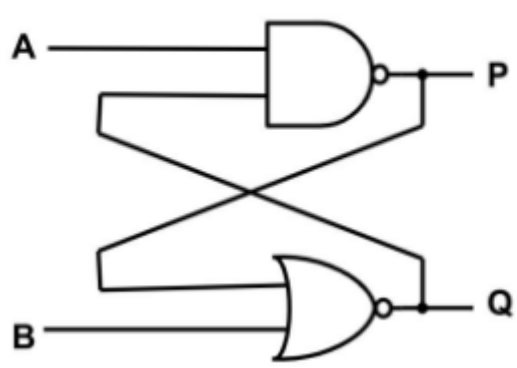
the essential prime implicants are _____.

A. $BD, \bar{B}\bar{D}$
B. BD, AB
C. $AB, \bar{B}\bar{D}$
D. $BD, \bar{B}\bar{D}, AB$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #49 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Consider the following latch-like circuit which uses both NAND and NOR gates. On the right side, you are provided with a sequence of changes of the inputs A and B. By filling in the blanks, you will write the output values P and Q for each step of the changes. For each blank, the value you fill in could be one of the following: 0,1 , or X (indeterminate value).



Initially, A = 0, B = 0, now P = _____ , Q = _____

then flip B, i.e., A = 0, B = 1, now P = _____ , Q = _____

then flip A, i.e., A = 1, B = 1, now P = _____ , Q = _____

then flip B, i.e, A = 1, B = 0, now P = _____ , Q = _____

What is the correct sequence of output values PQ for these changes?

A. $10 \rightarrow 10 \rightarrow 01 \rightarrow 01$
B. $10 \rightarrow 10 \rightarrow 10 \rightarrow \mathrm{XX}$
C. $10 \rightarrow 10 \rightarrow 10 \rightarrow 10$
D. $10 \rightarrow 10 \rightarrow 10 \rightarrow 01$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #50 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

For automata M, let $\langle \mathrm{M} \rangle$ denote some unary encoding of M, and $\mathrm{L}(\mathrm{M})$ denote the set of strings, over $\{0,1\}$, accepted by M.
Which of the following languages are Turing-decidable?

A. $\{\langle \mathrm{M} \rangle | \mathrm{M}$ is a deterministic Turing machine and M accepts $010\}$.
B. $\{\langle \mathrm{M} \rangle | \mathrm{M}$ is a nondeterministic Turing machine and M accepts $010\}$.
C. $\{\langle \mathrm{M} \rangle | \mathrm{M}$ is a DFA and M does not accept $010\}$.
D. $\{\langle \mathrm{M} \rangle \mid \mathrm{M}$ is a Turing machine and $\mathrm{L}(\mathrm{M}) = \Sigma^*\}$.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #51 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Let $\mathrm{M} = (\Sigma, Q, s, A, \delta)$ and $\mathrm{M}' = (\Sigma, Q, s, Q \backslash A, \delta)$ be arbitrary NFAs with identical alphabets, states, starting states, and transition functions, but with complementary accepting states.

M' is identical to M, except that accepting states in M are non-accepting in M' and vice versa.

Which of the following is/are viable possibilities (i.e. possible to be true for some NFA M) for $\mathrm{L}(\mathrm{M}), \mathrm{L}\left(\mathrm{M}'\right)$?

A. $\mathrm{L}(\mathrm{M}) = \mathrm{L}\left(\mathrm{M}'\right)$
B. $\mathrm{L}\left(\mathrm{M}'\right)$ is the complement of $\mathrm{L}(\mathrm{M})$
C. $\mathrm{L}(\mathrm{M}), \mathrm{L}\left(\mathrm{M}'\right)$ are neither same nor complementary.
D. $\mathrm{L}(\mathrm{M}), \mathrm{L}\left(\mathrm{M}'\right)$ are disjoint but not complementary.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52 | Numerical Type | Award: 2 | Penalty: 0 | Theory of Computation

Consider the regular language $\mathrm{L} = (11 + 11111)^*$. The minimum number of states in any DFA accepting this language is _____

Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #53 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Compiler Design

Consider the productions $\mathrm{A} \rightarrow \mathrm{PQ}$ and $\mathrm{A} \rightarrow \mathrm{XY}$. Each of the five non-terminals $\mathrm{A}, \mathrm{P}, \mathrm{Q}, \mathrm{X}$, and Y has two attributes: s is a synthesized attribute, and i is an inherited attribute. Consider the following rules.

- Rule $1: \mathrm{P}.i = \mathrm{A}.i + 2, \mathrm{Q}.i = \mathrm{P}.i + \mathrm{A}.s$ and $\mathrm{A}.s = \mathrm{P}.s + \mathrm{Q}.s$
- Rule $2: \mathrm{X}.i = \mathrm{A}.i + \mathrm{Y}.s$ and $\mathrm{Y}.i = \mathrm{X}.s + \mathrm{A}.i$

Which one of the following is TRUE?

A. Both Rule 1 and Rule 2 are L-attributed.
B. Only Rule 1 is L-attributed.
C. Only Rule 2 is L-attributed.

D. Neither Rule 1 nor Rule 2 is L-attributed.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #54 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

It is impossible to represent which of the following in a relational schema?

A. any mandatory participation constraint in a many-to-one relationship
B. any mandatory participation constraint in a many-to-many relationship
C. a one-to-one relationship
D. a many-to-one relationship

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #55 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Compiler Design

Calculate the liveness IN set for the following basic block, where the OUT set is calculated as $\{\mathrm{r3}, \mathrm{r4}\}$.

```
1. r5 =  &A + 12
2. r1 = load(r5)
3. r3 = r1 + r2
```

OUT = $\{\mathrm{r3}, \mathrm{r4}\}$

A. $\mathrm{IN} = \{r1, r2, r4\}$
B. $\mathrm{IN} = \{\mathrm{r2}, \mathrm{r4}\}$
C. $\mathrm{IN} = \{r2, r3, r4\}$
D. $\mathrm{IN} = \{r1, r2\}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

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